

RETAIL SALES FORECASTING

A Comparative Study of Statistical and Machine Learning Methods for Store–Department Weekly Sales Forecasting

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Executive Summary

Business problem. A multi-store retail chain requires accurate weekly sales forecasts at the Store–Department level to guide inventory planning, staffing allocation, and promotional decision-making. Forecast error in either direction is costly: over-forecasting ties up capital in excess inventory, while under-forecasting drives stock-outs and lost sales.

Approach. Company-wide weekly sales were aggregated into a single series and modelled using classical statistical methods (ARIMA, SARIMA) to characterise trend and seasonality at the business level. In parallel, four machine learning models — Linear Regression, Decision Tree, Random Forest, and XGBoost — were trained at full Store–Department–week granularity using lag and rolling-window features, calendar and holiday indicators, and store attributes. A strict, chronological train/validation/test split was used throughout to prevent data leakage.

Key result. The **Random Forest** model produced the strongest and most reliable granular forecasts (MAE $\approx$ \$1,226.53; weighted MAE $\approx$ \$1,281.71; $R^2 = 0.987$), outperforming Linear Regression, Decision Tree, and XGBoost across every error metric evaluated. At the aggregate company level, **SARIMA(0,0,1)(0,1,1,52)** was the stronger statistical model, achieving a MAPE of 2.67% against 3.30% for ARIMA, by explicitly representing the 52-week seasonal cycle inherent to retail demand.

Key drivers. Across all three tree-based models, the strongest predictor of weekly sales is the same week's sales one year earlier (*lag_52*), confirming that year-over-year seasonality outweighs short-term trend and macroeconomic variables as a forecasting signal. Holiday effects are material but concentrated: the week before Christmas and the Thanksgiving week show the largest uplifts (+66.6% and +39.7% respectively), while the calendar week nominally flagged as “Christmas” is comparatively muted.

Recommendation. Deploy the Random Forest model for operational Store–Department demand planning, align promotional and staffing calendars with the identified high-lift weeks rather than generic holiday flags, and retain the SARIMA model as a lightweight, aggregate-level check on company-wide trend.

Headline Metrics

Best granular model: Random Forest — MAE \$1,226.53 | WMAE \$1,281.71 | R^2 0.987

Best aggregate model: SARIMA(0,0,1)(0,1,1,52) — MAPE 2.67% | RMSE 3.36% of mean sales

Dataset: 420,212 weekly records | 45 stores | 81 departments | Feb 2010 – Oct 2012

Top predictive driver: *lag_52* (same week, prior year) — dominant across all tree-based models

1. Introduction

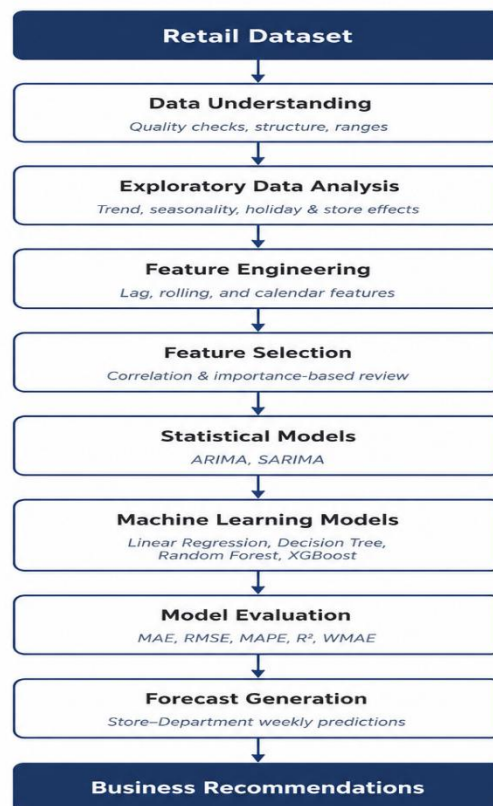
1.1 Business Problem and Objective

A retail chain operating 45 stores and 81 departments needs to forecast weekly sales for every Store–Department combination. Reliable forecasts at this granularity directly support three operational decisions: how much inventory to stock, how to allocate staff hours, and when to schedule promotions. The chain's existing planning process relies on simple historical averaging, which does not adequately capture seasonal peaks, holiday effects, or the interaction between promotional markdowns and demand.

Objective: develop and validate an end-to-end forecasting pipeline that combines classical statistical time series methods with machine learning, and identify the model and feature set that yield the most accurate and operationally useful forecasts.

1.2 Project Workflow

The analysis followed a structured, ten-stage pipeline, from raw data through to actionable business recommendations, illustrated below.



2. Dataset Description

Attribute	Value
Records	420,212
Columns	24
Stores	45
Departments	81
Unique Store–Department series	3,323 (of 3,645 possible combinations)
Time period	Feb 2010 – Oct 2012 (143 weeks)
Target variable	Weekly_Sales
Missing values	None
Duplicate records	None

The dataset arrived pre-cleaned: no missing values were present in any column, and no duplicate records or duplicate Store–Department–Date combinations were found. The Date field was parsed to a proper datetime type, and the data was sorted by Store, Department, and Date to preserve chronological order — a prerequisite for correct lag-feature construction and leakage-free evaluation.

2.1 Tools & Technologies

Tool / Library	Purpose
Python	Programming language used for analysis and modelling
Pandas	Data cleaning, manipulation, and feature engineering
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical visualizations
Scikit-learn	Machine learning models and evaluation
Statsmodels	ARIMA and SARIMA forecasting
XGBoost	Gradient boosting model
Jupyter Notebook	Development environment

3. Research Methodology

The forecasting pipeline followed seven sequential stages, each building on the output of the previous one:

- **Data Understanding** — structural review, quality checks, and range validation of the raw dataset.
- **Exploratory Data Analysis** — characterising trend, seasonality, and the influence of stores, departments, holidays, and promotions.
- **Feature Engineering** — constructing lag, rolling-window, and calendar features within each Store–Department series.
- **Feature Selection** — reviewing correlation structure and, later, fitted-model importances to finalise the feature set.
- **Statistical Models** — ARIMA and SARIMA, fitted to the aggregated company-wide series.
- **Machine Learning Models** — Linear Regression, Decision Tree, Random Forest, and XGBoost, fitted at full Store–Department granularity.
- **Evaluation and Forecast Generation** — comparing all models on a held-out test set and generating final Store–Department forecasts from the selected model.

4. Exploratory Data Analysis

Exploratory analysis was conducted to characterise the distribution, trend, seasonality, and key drivers of weekly sales prior to feature engineering and modelling.

4.1 Distribution of Weekly Sales

Weekly sales are strongly right-skewed, with a small number of very large values pulling the distribution away from a normal shape. The maximum observed weekly sales value is \$693,099, far above the 75th percentile of approximately \$20,271, confirming the presence of substantial outliers driven by high-volume Store–Department combinations and peak sales periods.

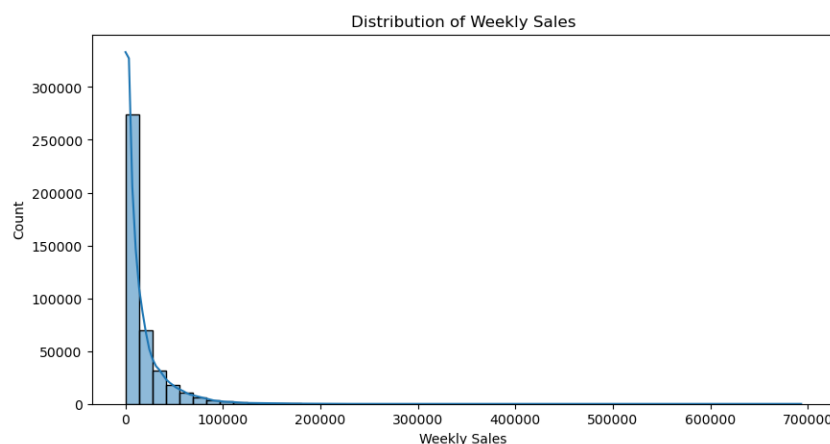


Figure 1. *Distribution of weekly sales across all Store–Department–week records, showing a pronounced right skew.*

4.2 Trend and Seasonality

Aggregating sales across all stores and departments by date reveals clear recurring peaks corresponding to major shopping periods, with sales otherwise relatively stable and exhibiting a slight long-term upward trend.

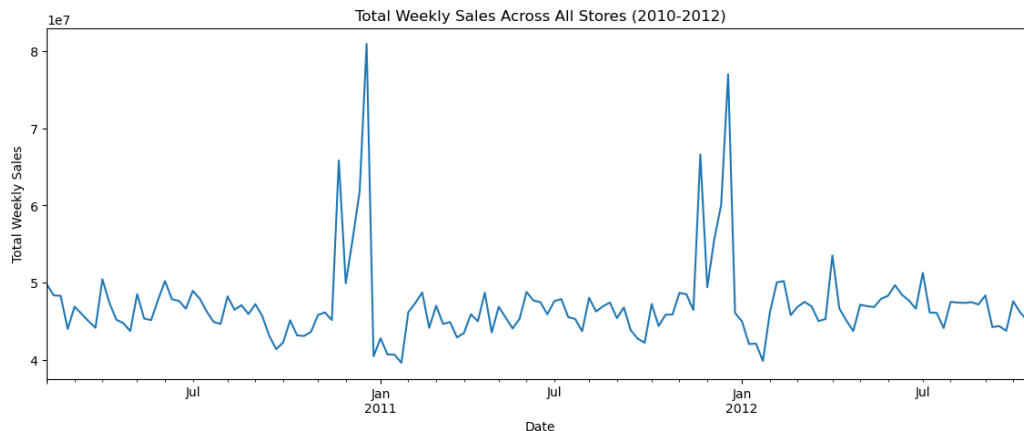


Figure 2. Total weekly sales aggregated across all stores, 2010–2012.

A seasonal decomposition (additive, 52-week period) of the aggregated series confirms this structure quantitatively: the trend component declines slightly in the early period before rising gradually from mid-2011 onward, and the seasonal component shows clear recurring peaks at the same point in each annual cycle.

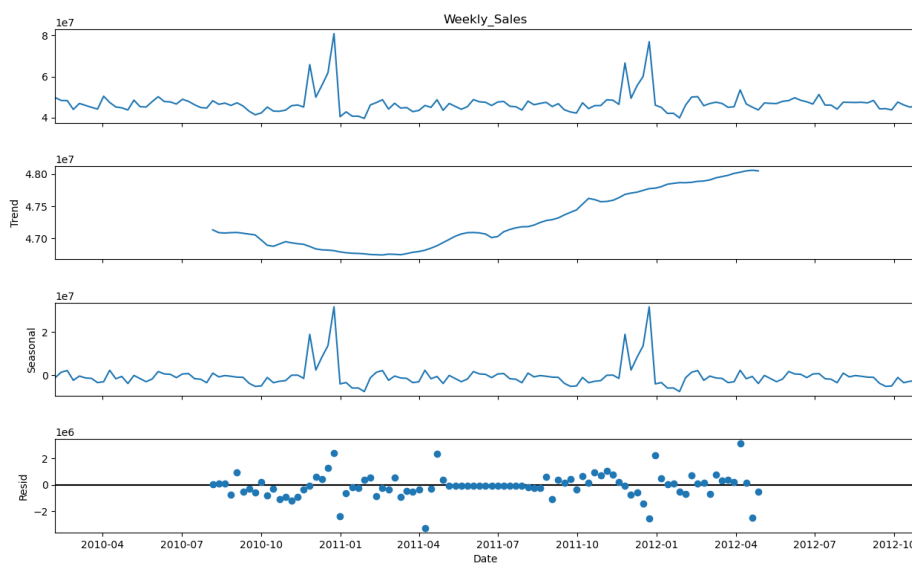


Figure 3. Additive seasonal decomposition of aggregate weekly sales into trend, seasonal, and residual components.

4.3 Store and Department Effects

Sales vary substantially across both stores and departments. Store type is also informative: Type A stores generate the highest average weekly sales (\$20,148) and are on average the largest by floor size (182,225 sq. ft.), followed by Type B (\$12,291) and Type C (\$9,549).

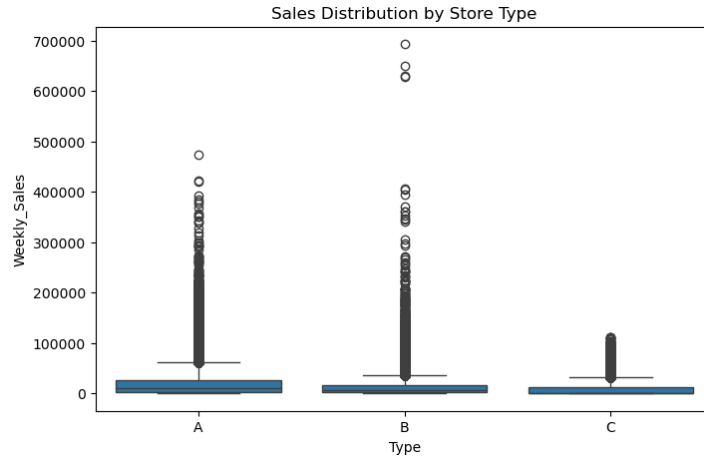


Figure 4. Distribution of weekly sales by store type (A, B, C).

4.4 Holiday and Promotional Effects

Average weekly sales are 7.2% higher during flagged holiday weeks than non-holiday weeks, but this masks large variation by specific holiday:

Holiday / Period	Change vs. Normal Week
Thanksgiving	+39.7%
Week before Christmas (week 51)	+66.6%
Super Bowl	+2.5%
Labor Day	-0.6%
Christmas (flagged week)	-8.8%

Notable finding: the week immediately preceding Christmas drives substantially higher sales than the calendar week officially flagged as “Christmas.” This observation motivated a dedicated *week_before_christmas* feature, since the standard holiday flag alone would understate the true seasonal peak.

Promotional markdowns show a smaller, time-dependent effect: sales increase by 1.9% overall on markdown weeks, rising to 4.8% when restricted to 2011–2012 (the period after markdowns were introduced). Year is a confounding variable here, since both markdown adoption and baseline sales rise together over time.

4.5 Correlation Analysis

A correlation heatmap across numeric fields shows that most variables — temperature, fuel price, CPI, unemployment — have only weak linear correlation with weekly sales, while store size and the markdown variables show moderate mutual correlation. No single variable dominates linearly, suggesting non-linear, tree-based models are better suited to capturing the underlying relationships than a purely linear approach.

4.6 EDA Summary

- Weekly sales exhibit strong annual seasonality and recurring holiday peaks.
- Sales vary considerably across stores and departments; store type and size are informative.
- Thanksgiving and the week before Christmas produce the largest holiday-driven sales increases.
- Promotional markdowns contribute positively but are concentrated in later years, confounding a simple before/after comparison.
- Linear correlations with weekly sales are generally weak, favouring non-linear machine learning models.

5. Feature Engineering

Feature engineering translated the EDA findings into model-ready variables, spanning calendar features and lag/rolling statistics computed within each Store–Department series. Because lag and rolling features require historical observations at the start of every series, rows without sufficient history were dropped rather than imputed, reducing the modelling dataset from 420,212 to 260,298 rows (approximately 62% retained) — a deliberate trade-off that preserves the integrity of the lag features.

Feature	Purpose
Week	Weekly seasonality
Month	Monthly trend
Quarter	Quarterly pattern
Year	Long-term trend
Lag_1	Previous week's sales
Lag_2	Short-term dependency
Lag_4	Monthly behaviour
Lag_13	Quarterly seasonality
Lag_26	Half-year trend
Lag_52	Previous year's same week
Rolling Mean (4-week)	Recent demand trend
Rolling Std (4-week)	Sales volatility
is_peak_week / week_before_christmas	Captures the true holiday sales peaks identified in EDA

Note - Store Type (A/B/C) was one-hot encoded with the first category dropped, rather than encoded ordinally, to avoid imposing an artificial numeric ordering between store categories that does not reflect a real ranking.

5.1 Data Splitting Strategy

- **Statistical Models (ARIMA, SARIMA):** The aggregated weekly sales series was divided into 80% training and 20% testing data.
- **Machine Learning Models:** The engineered Store–Department dataset was divided into 70% training, 15% validation, and 15% testing. The validation set was used exclusively for hyperparameter tuning of XGBoost, while the test set remained completely unseen until final evaluation.

This chronological splitting strategy closely reflects real-world forecasting, where future sales are predicted using only historical observations.

6. Feature Selection

Feature selection was performed to identify the most relevant predictors while ensuring that redundant variables did not adversely affect model performance. This involved two stages: correlation analysis to assess multicollinearity and feature importance analysis using tree-based machine learning models.

6.1 Correlation Analysis

The correlation matrix was used to examine relationships among the engineered numerical features. Most variables exhibited weak to moderate correlations, while stronger correlations were observed mainly among lag features and rolling statistics, as these are derived from historical sales.

Although lag-based features are correlated, they capture demand patterns at different time horizons and therefore provide complementary information. No severe multicollinearity was identified that required feature removal.

Key Findings

- Most predictors show weak to moderate correlations.
- Higher correlations are limited to lag and rolling features.
- No feature was removed based on correlation analysis.

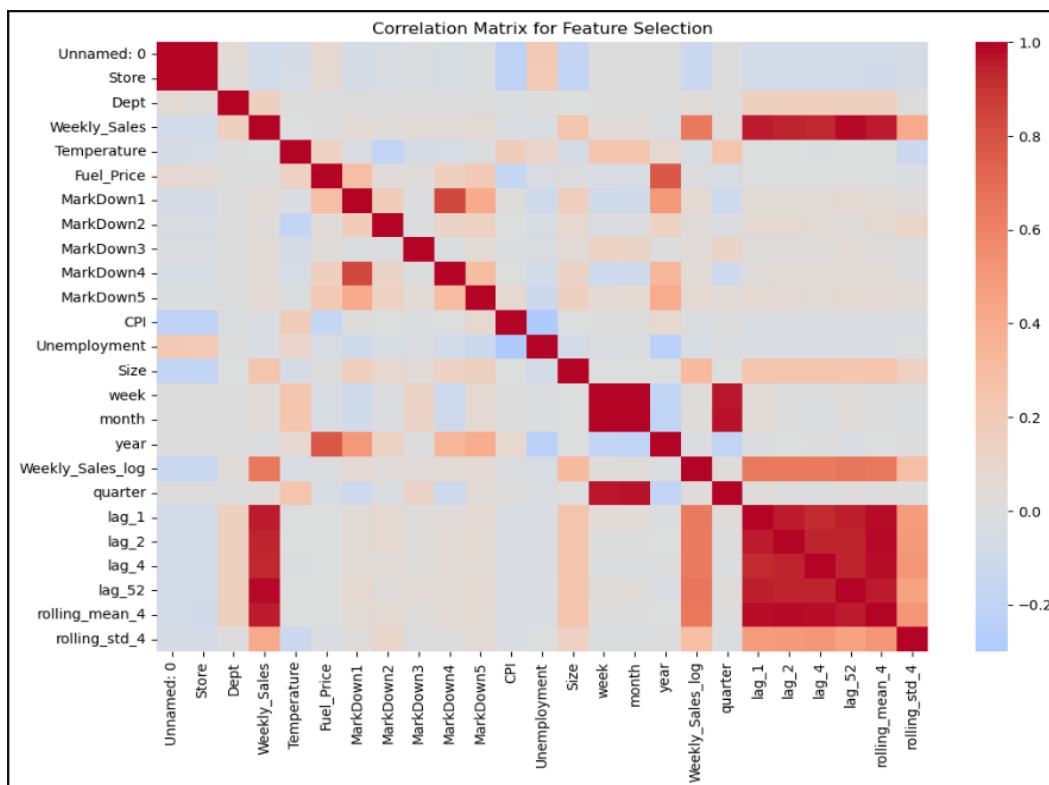


Figure 5. Correlation Matrix for Feature Selection

6.2 Feature Importance Analysis

Feature importance scores from the Decision Tree, Random Forest, and XGBoost models were averaged to obtain an overall ranking (Figure 6.2). The results show that lag_52 is the most influential feature, followed by lag_1, rolling_mean_4, and other lag variables, highlighting the importance of historical sales patterns in forecasting.

Store characteristics, promotional variables, and holiday indicators contributed comparatively less but still provided useful contextual information.

Key Findings

- lag_52 is the most important predictor.
- Historical sales features dominate the forecasting task.
- Feature rankings are consistent across all tree-based models.

6.3 Feature Selection Decision

Based on the combined findings from correlation analysis and feature importance analysis, no variables were removed from the final feature set.

Correlation analysis confirmed that the engineered predictors did not exhibit problematic multicollinearity, while feature importance demonstrated that lag variables, rolling statistics, calendar features, store attributes, and economic indicators each contributed useful predictive information. Because ensemble tree models effectively manage correlated inputs, removing features would have provided little benefit and could have reduced predictive performance.

Final Feature Set

- **Calendar Features** (Week, Month, Quarter, Year)
- **Lag Features** (Lag_1, Lag_2, Lag_4, Lag_13, Lag_26, Lag_52)
- **Rolling Statistics** (Rolling Mean, Rolling Standard Deviation)
- **Holiday Indicators**
- **Store Characteristics** (Store, Department, Type, Size)
- **Economic Variables** (Temperature, Fuel Price, CPI, Unemployment)
- **Promotional Variables** (Markdown1–Markdown5)

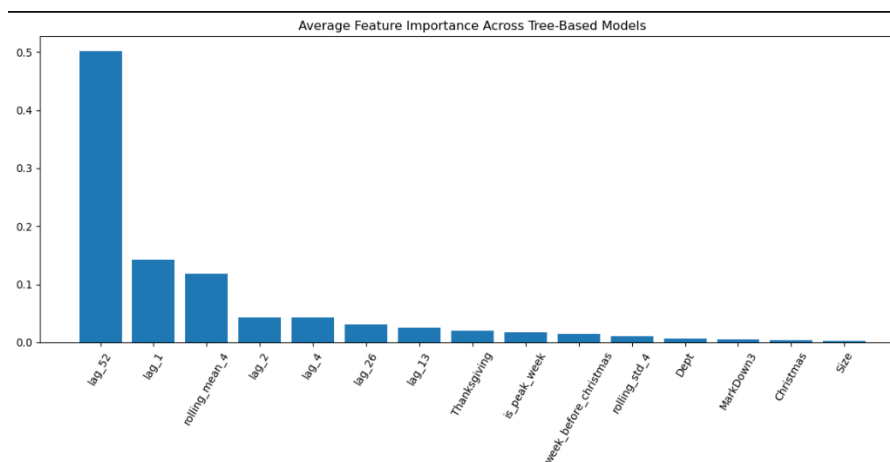


Figure 6: Average Feature Importance Across Tree-Based Models

7. Statistical Forecasting

7.1 Why Aggregation Was Required

The retail dataset is a panel: multiple Store–Department observations exist for each calendar week. Classical univariate methods such as ARIMA and SARIMA require a single observation per time step, so weekly sales were summed across all stores and departments to produce one aggregated, company-wide series before applying these models. The aggregated series (143 weekly observations) was split chronologically into 114 weeks of training data and 29 weeks of test data.

7.2 ARIMA

Candidate autoregressive and moving-average orders were identified from the autocorrelation structure of the training series and compared by AIC/BIC. The best-performing specification was ARIMA(2,0,2).

ARIMA captures the broad trend in aggregate weekly sales but does not explicitly model the recurring annual seasonal pattern, limiting its forecasting accuracy relative to a seasonal alternative.

7.3 SARIMA

A grid search over seasonal and non-seasonal orders was conducted, ranked by AIC. The selected specification, SARIMA(0,0,1)(0,1,1,52), explicitly models the 52-week seasonal cycle present in the data.

Residual diagnostics indicate mild remaining autocorrelation and a small positive bias (a slight tendency to under-forecast), attributable in part to having only about 2.2 full seasonal cycles of training history available for a 52-week seasonal model. This does not undermine the model's strong demonstrated out-of-sample accuracy, validated directly against held-out test data; adding holiday indicators as exogenous regressors is noted as a natural refinement.

7.4 Comparison and Interpretation

Model	MAE	RMSE	% RMSE	MAPE
ARIMA(2,0,2)	1,528,077.22	1,952,054.85	4.18%	3.30%
SARIMA(0,0,1)(0,1,1,52)	1,254,363.66	1,570,347.85	3.36%	2.67%

SARIMA outperforms ARIMA across every metric, as expected given that it explicitly represents the seasonal component present in weekly retail sales, while ARIMA captures only non-seasonal trend behaviour. Neither statistical model, however, incorporates explanatory variables such as store characteristics, department information, promotions, or holidays — which motivates the machine learning approach that follows.

SARIMA explicitly models the yearly seasonal cycle using seasonal differencing and seasonal moving-average terms. Since retail sales exhibit recurring annual demand patterns, SARIMA captures these fluctuations more effectively than ARIMA, resulting in lower forecasting errors.

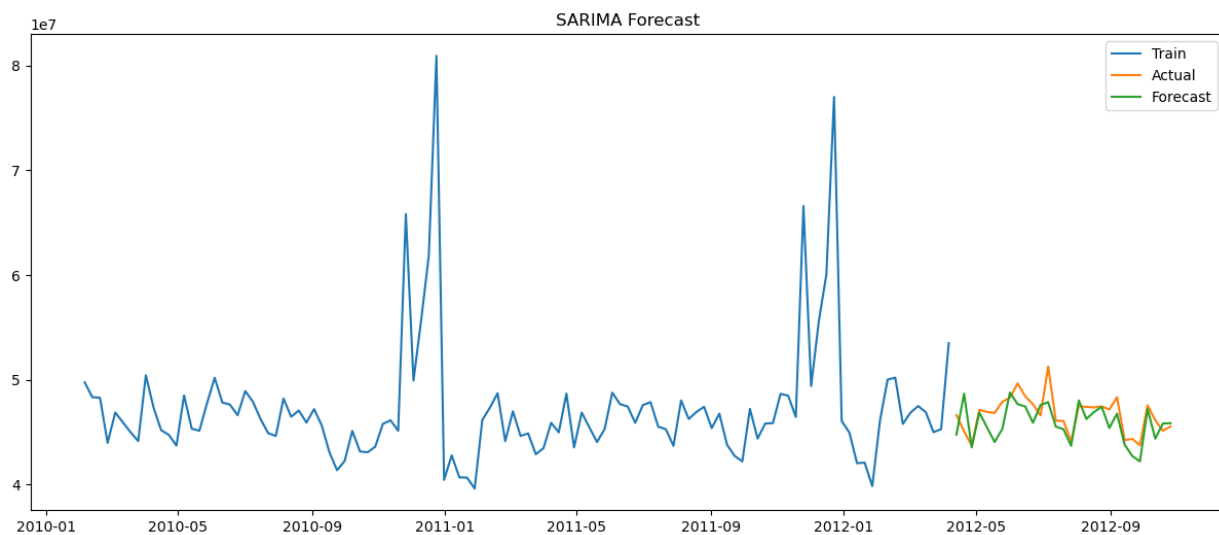


Figure 7. SARIMA(0,0,1)(0,1,1,52) forecast against the held-out test period.

8. Machine Learning Models

While ARIMA and SARIMA capture temporal trend and seasonality well at the aggregate level, they cannot exploit store characteristics, department identity, holiday indicators, promotional markdowns, or macroeconomic factors at the Store–Department grain. Machine learning models were therefore developed to learn these more complex, non-linear relationships directly at full granularity, using a chronological 70% / 15% / 15% - train / validation / test split (validation reserved exclusively for XGBoost tuning, keeping the test set untouched until final evaluation).

8.1 Linear Regression

A standard linear regression was fitted as an interpretable baseline against which the non-linear models are compared.

8.2 Decision Tree

A single Decision Tree Regressor (max depth 15, minimum samples per split 10, minimum samples per leaf 5) was fitted to capture non-linear structure.

Feature importance is highly concentrated in this model: lag_52 alone accounts for 81.1% of total importance, with lag_1 (13.9%) and rolling_mean_4 (3.2%) contributing most of the remainder — a structural property of single decision trees, which greedily select the single most informative split at each node.

8.3 Random Forest — Selected Model

A Random Forest Regressor (200 trees, max depth 15, minimum samples per split 10, minimum samples per leaf 5, sqrt feature subsampling) was fitted next.

Random Forest achieves the lowest MAE and RMSE and the highest R^2 among all four models. Its per-split random feature subsampling spreads importance more evenly across lag_52, lag_1, rolling_mean_4, lag_2, and lag_26 rather than concentrating on a single feature, which is a key reason it generalises better than the single Decision Tree on this data.

8.4 XGBoost and Hyperparameter Tuning

An XGBoost Regressor was tuned via RandomizedSearchCV over 30 candidate parameter settings (n_estimators, max_depth, learning_rate, subsample, colsample_bytree), using 3-fold TimeSeriesSplit cross-validation — never standard K-Fold — to preserve chronological order and avoid leakage, scored on negative MAE. The search evaluated 90 total model fits; the best configuration found was 600 estimators, max depth 10, learning rate 0.05, subsample 1.0, and column-subsample 0.8.

Despite systematic tuning, XGBoost did not surpass the untuned Random Forest on this dataset, though it remains competitive and outperforms both Linear Regression and the single Decision Tree.

8.5 Residual Analysis

Residual diagnostics for the leading model (Random Forest) show errors distributed closely around zero with no strong systematic bias, and no obvious pattern of increasing error at higher predicted sales values — indicating consistent accuracy across the sales range rather than degradation at scale.

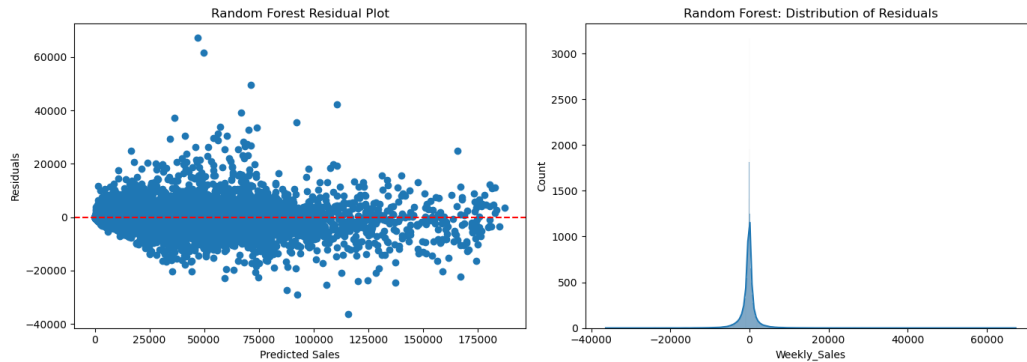


Figure 8. Random Forest residual diagnostics: residuals vs. predicted values, and residual distribution.

9. Model Comparison

All four machine learning models were compared on the same held-out test set using MAE, RMSE, MAPE-derived percentage error, holiday-weighted MAE (WMAE, which up-weights holiday and peak weeks 5× to reflect their greater business importance), and R^2 .

Model	MAE	% MAE	RMSE	% RMSE	WMAE	R^2
Random Forest	1,226.53	7.70%	2,496.83	15.68%	1,281.71	0.987
XGBoost (Tuned)	1,339.64	8.41%	2,810.39	17.65%	1,375.25	0.984
Decision Tree	1,489.07	9.35%	3,179.38	19.97%	1,524.73	0.979
Linear Regression	1,613.06	10.13%	3,007.43	18.89%	1,644.71	0.981

Selection: Random Forest is selected as the final forecasting model. It achieves the lowest MAE, RMSE, and holiday-weighted WMAE, and the highest R^2 , across all four evaluated models — a consistent ranking across both plain and holiday-weighted error measures gives confidence that the result is not an artefact of any single metric.

10. Forecast Generation

The selected Random Forest model was used to generate weekly Store–Department sales predictions across the full held-out test period. For every test record, the output includes actual sales, predicted sales, forecast error, and absolute percentage error, providing both record-level detail and a basis for aggregate evaluation.

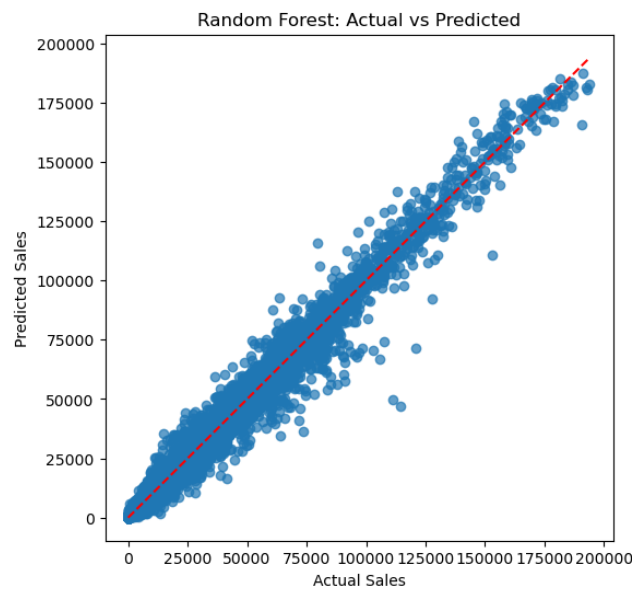


Figure 9. Actual vs. predicted sales across the full test set. Points close to the diagonal reference line indicate accurate forecasts.

Most predictions cluster tightly around the diagonal reference line, indicating consistently accurate predictions across the sales range, including for larger, higher-variance Store–Department series.

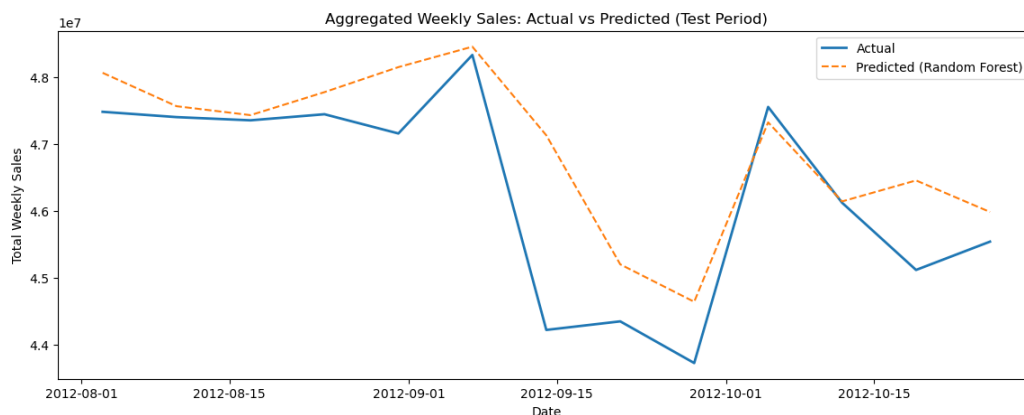


Figure 10. Aggregated actual vs. predicted weekly sales, summed across all stores and departments, over the test period.

Summing predictions across all stores and departments for each test-period week shows that the model tracks the aggregate sales pattern closely — a useful sign that granular Store–Department accuracy also rolls up correctly to company-level planning.

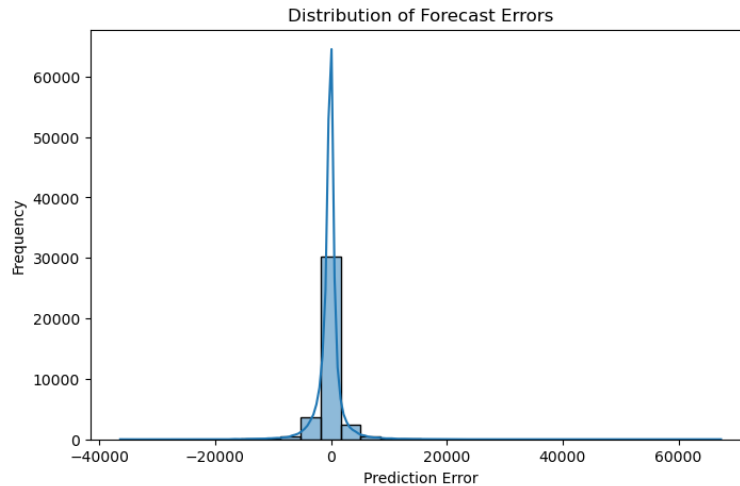


Figure 11. Distribution of forecast errors (actual minus predicted) across the test set.

The error distribution is centred close to zero and roughly symmetric, indicating no strong systematic tendency toward over- or under-forecasting at the aggregate level.

Sample Forecast Output

Date	Actual Sales	Predicted Sales	Error	APE %
2012-08-03	16,628.31	17,433.32	-805.01	4.84%
2012-08-10	16,119.92	16,992.99	-873.07	5.42%
2012-08-17	17,330.70	16,235.85	1,094.85	6.32%
2012-08-24	16,286.40	17,489.09	-1,202.69	7.38%
2012-09-07	18,322.37	17,886.14	436.23	2.38%
2012-10-05	21,904.47	21,653.71	250.76	1.14%

Forecast Output File

The complete forecast results have been attached as an Excel workbook for reference. The file includes actual sales, predicted sales, forecast errors, and percentage errors for the test period.



11. Business Insights

11.1 Which Factors Impact Sales Most?

lag_52 — sales in the same week one year earlier — is the top-ranked feature across all three tree-based models, but the degree of reliance differs sharply: Decision Tree concentrates 81% of its importance there alone, while Random Forest and XGBoost spread importance more evenly across lag_52, lag_1, rolling_mean_4, lag_2, and lag_26. This is expected, since Random Forest's per-split feature subsampling deliberately prevents any single feature from dominating every tree. Either way, engineered lag and rolling features far outweigh macroeconomic factors such as CPI, unemployment, or fuel price.

The dominance of lag-based features suggests that historical sales behaviour is a stronger predictor than external macroeconomic variables, indicating that operational planning should primarily rely on historical demand patterns while using economic indicators as supplementary information.

11.2 How Do Holidays Affect Sales?

Holiday weeks average 7.2% higher sales than non-holiday weeks. More importantly, the week before Christmas (+66.6%) and Thanksgiving (+39.7%) generate the strongest increases — substantially larger than the calendar week officially flagged as “Christmas,” which is comparatively muted.

11.3 Do Promotional Markdowns Drive Sales?

Markdowns show a modest positive association with sales, but year is a confounding variable, since both markdown adoption and baseline sales rise together over time. Feature importance analysis confirms markdown variables contribute relatively little individually compared to lag and calendar features.

11.4 Which Stores and Store Types Perform Best?

Type A stores generate the highest average weekly sales and are, on average, the largest by floor size; store size shows a consistent positive relationship with weekly sales. Sales are also concentrated: a small number of stores and departments account for a disproportionate share of total revenue.

12. Recommendations

1. Anchor demand planning on year-over-year seasonality

Use lag_52 and year-over-year seasonal patterns as the primary basis for demand planning, with short-term trend applied only as a secondary adjustment.

2. Re-time holiday inventory and staffing around true peak weeks

Build inventory and staffing ahead of the week before Christmas and Thanksgiving week specifically, rather than relying on standard holiday-week flags, which understate or mis-time the real peak.

3. Validate markdown effectiveness with a causal design

Evaluate markdown effectiveness using a design that separates the promotional effect from the underlying time trend (e.g. difference-in-differences) before expanding markdown spend on the basis of the current raw effect size.

4. Prioritise forecasting accuracy where the dollars are

Focus forecast monitoring and inventory buffers on the highest-volume stores and departments, where forecast error carries the largest absolute dollar impact.

5. Deploy Random Forest, with SARIMA as a company-wide check

Use the Random Forest model for day-to-day Store–Department forecasting, and the SARIMA model as a lightweight, complementary check on total company trend.

13. Limitations and Future Work

- The statistical-model train/test split does not include the November–December holiday peak, so ARIMA/SARIMA accuracy during major holiday spikes was not directly validated by that split.
- SARIMA training used only approximately 2.2 full seasonal cycles (114 weeks) for a 52-week seasonal model, producing reduced confidence in individual coefficient significance, although out-of-sample forecast accuracy was validated directly against held-out data.
- MAPE is sensitive to near-zero actual sales values in a subset of Store–Department–weeks and can overstate error at the granular ML level; MAE and MAE-as-percent-of-mean are reported as the more representative accuracy measures for the ML models.
- Rows without sufficient lag history were dropped rather than imputed, reducing the ML modelling dataset from 420,212 to 260,298 rows (~62% retained).
- Promotional markdown effects are confounded with the overall time trend in the comparison presented and would benefit from a dedicated causal-inference approach.
- XGBoost was tuned via randomised search (30 candidates $\times$ 3 time-series folds); a larger or exhaustive search may yield further improvement but was not pursued given the strong performance already achieved by Random Forest.
- Future work should pilot the Random Forest forecasts in one region for a full seasonal cycle, re-validate specifically ahead of the November–December peak, and refresh the models periodically as new seasons of data become available.

14. Conclusion

An end-to-end retail sales forecasting pipeline was developed and validated, combining classical statistical time series methods with machine learning across six candidate models: ARIMA, SARIMA, Linear Regression, Decision Tree, Random Forest, and XGBoost.

At the aggregate company level, SARIMA outperformed ARIMA by explicitly modelling the 52-week seasonal cycle present in retail demand. At the operational Store–Department level, Random Forest achieved the best predictive performance among the machine learning models effectively capturing non-linear relationships between historical sales, holidays, markdowns, and store characteristics that the linear and single-tree alternatives could not fully exploit.

This project demonstrates that combining statistical time-series analysis with machine learning provides a robust framework for retail demand forecasting. Among all evaluated models, Random Forest achieved the highest forecasting accuracy and is recommended for operational deployment.